

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1804

West -Station 138 kV
July 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1804. The Interconnection Customer proposes a 95.5 MW solar photovoltaic facility interconnecting at the West -Station 138 kV substation in the Maryland area. The request was received on July 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 644 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,710,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dobbins to Spencer Substation 138 kV	Easton 2 to Vienna Operatons 345 kV	183.0	494	6.0
Easton to NavyMole 138 kV	Easton 2 to Vienna Oportions 345 kV line	139.3	609	6.0
Dobbins to Easton 138 kV	Vienna Operations to Easton 2 345 kV	139.2	636	6.0
Bozman to Greenstone Substation 138 kV	Easton 2 to Cady 345 kV	138.4	286	10.8
Foreman Cement to Greenstone Substation 138 kV	Cady to Easton 2 345 kV	130.8	295	10.8
Wye Mills to Grasoville 138 kV	Easton 2 to Cady 345 kV	121.4	303	5.8
Foreman Cement to Whispering Pines 138 kV	Easton 2 to Cady 345 kV line	117.8	313	10.8
Wheatland Sta to Mckee Run 138 kV	Cady to Dodd Road 345 kV line	116.1	311	6.1
Park Ave to Harbor 138 kV L	Cady to Easton 2 345 kV Line	115.6	589	9.5
Holston Dam to Harbor 138 kV	Cady to Easton 2 345 kV	114.6	611	9.5
Dell Crk Substation to Churchs Ferry 138 kV	Dodd Rd to Cady Substation 345 kV	110.8	225	5.6
Stave Falls to C W Tippy 138 kV	Cady to Dodd Rd Substation 345 kV line	110.6	312	8.7
Grasonville to Macon 138 kV	Cady to Easton 2 345 kV L	108.8	615	11.3
Wye Mills to Holston Dam 138 kV L	Cady to Easton 2 345 kV	107.1	348	5.8
Holston Dam Substation to Carancahua 138 kV L	Cady to Easton 2 345 kV	106.7	333	5.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Stave Falls to Stherland Springs 138 kV	Dodd Rd to Cady 345 kV	106.1	316	8.7
Hillsboro to Park Ave 138 kV line	Cady to Easton 2 345 kV	105.8	1,143	25.0
Limon to Clatskanie 138 kV Line	Easton 2 to Cady 345 kV Line	105.3	243	6.4
Cady to Dodd Road 345 kV	Bond-Elk Grove Florin 345-138 kV Transformer	104.3	2,028	57.7
Holston Dam to Navy Mole 138 kV L	Cady to Easton 2 345 kV	103.7	1,399	20.8
Price to White Bluff Substation 138 kV	Cady to Easton 2 345 kV	103.0	613	11.3
Lost Hills to Oak Hill 138 kV	Cady to Dodd Rd 345 kV	102.8	655	14.4
Warren F Sam Beasley Generation Sta to Wheatland Station 138 kV	Dodd Road to Cady 345 kV L	102.5	267	5.5
Limon Substation to Vernon 138 kV	Cady to Easton 2 345 kV	101.8	699	13.5
Calvert Cliffs 345/138kV TX	Cady Substation to Easton 2 345 kV	101.8	817	26.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dobbins to Spencer Substation 138 kV	Rebuild line with higher-capacity conductor	\$2,685,000	5% (\$124,000)
Easton to Nav Mole 138 kV	Rebuild line with higher-capacity conductor	\$8,393,000	5% (\$389,000)
Dobbins to Easton 138 kV	Rebuild line with higher-capacity conductor	\$2,177,000	5% (\$101,000)
Bozman to Greenstone Substation 138 kV	Rebuild line with higher-capacity conductor	\$38,120,000	2% (\$930,000)
Foreman Cement to Greenstone Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,407,000	2% (\$108,000)
Grasonville to Wye Mills 138 kV L	Rebuild line with higher-capacity conductor	\$12,660,000	2% (\$282,000)
Whispering Pines to Foreman Cement 138 kV	Rebuild line with higher-capacity conductor	\$16,858,000	2% (\$411,000)
Wheatland Sta to Mckee Run 138 kV	Rebuild line with higher-capacity conductor	\$2,752,000	2% (\$60,000)
Park Ave to Harbor 138 kV line	Rebuild line with higher-capacity conductor	\$22,128,000	2% (\$395,000)
Holston Dam to Harbor 138 kV L	Rebuild line with higher-capacity conductor	\$13,385,000	2% (\$239,000)
Dell Crk to Churchs Ferry 138 kV line	Rebuild line with higher-capacity conductor	\$2,136,000	2% (\$42,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
C W Tippy to Stave Falls 138 kV	Rebuild line with higher-capacity conductor	\$2,917,000	2% (\$60,000)
Grasonville to Macon 138 kV	Reconductor line	\$1,584,000	2% (\$34,000)
Wye Mills to Holston Dam 138 kV L	Reconductor line	\$3,890,000	2% (\$87,000)
Carancahua to Holston Dam 138 kV	Reconductor line	\$3,540,000	2% (\$79,000)
Stave Falls to Stherland Springs 138 kV	Reconductor line	\$1,082,000	2% (\$22,000)
Hillsboro to Park Ave 138 kV line	Reconductor line	\$2,485,000	2% (\$59,000)
Limon to Clatskanie 138 kV Line	Reconductor line	\$2,600,000	2% (\$55,000)
Dodd Rd to Cady 345 kV Line	Reconductor line	\$17,596,000	2% (\$380,000)
Holston Dam to Navy Mole 138 kV L	Reconductor line	\$1,065,000	2% (\$21,000)
White Bluff to Price 138 kV	Reconductor line	\$6,541,000	2% (\$141,000)
Lost Hills to Oak Hill 138 kV	Reconductor line	\$7,427,000	2% (\$167,000)
Warren F Sam Beasley Generation Sta to Wheatland Station 138 kV	Reconductor line	\$3,414,000	2% (\$82,000)
Veron Substation to Limon 138 kV L	Reconductor line	\$1,300,000	2% (\$30,000)
Calvert Cliffs 345-138 kV Transformer	Replace transformer with higher-rated unit	\$15,802,000	3% (\$412,000)
	Total allocated to project		\$4,710,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.